ANSWERS

1 **(a)** constant <u>phase difference</u> …………………………………………………………….. B1 [1]

(b) allow wavelength estimate 750 nm $\rightarrow$ 550 nm …………………………………….. C1
separation = $\lambda D / x$ …………………………………………………………………. C1
= $(650 \times 10^{-9} \times 2.4) / (0.86 \times 10^{-3})$
= 1.8 mm ……………………………………………………………………. A1 [3]
(*allow 2 marks from inappropriate estimate if answer is in range 10 cm → 0.1 mm)*

(c) no longer complete destructive interference /
amplitudes no longer completely cancel …………………………………………….. M1
so dark fringes are lighter ……………………………………………………………….. A1 [2]

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2 **(a)** when a <u>wave</u> (front) passes by/incident on an edge/slit ……………………… M1
wave bends/spreads (into the geometrical shadow) ……………………………… A1 [2]

(b) $\tan\theta = \dfrac{38}{165}$
$\theta = 13°$ ……………………………………………………………………………… C1
$d \sin\theta = n\lambda$ …………………………………………………………………………… C1
$d = 2.82 \times 10^{-6}$ ……………………………………………………………………………C1
number = $(1/d =)\ 3.6 \times 10^{5}$ ……………………………………………………………… A1 [4]

(c) P remains in same position ……………………………………………………… B1
X and Y rotate through 90° ……………………………………………………………. B1 [2]

(d) *either* screen not parallel to grating
or grating not normal to (incident) light …………………………………………. B1 [1]

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3 **(a)** when a wave passes through a slit / by an edge M1
the wave spreads out / changes direction A1 [2]

(b) diagram: wavelength unchanged M1
wavefront flat at centre, curving into geometrical shadow A1 [2]

(c) $d \sin\theta = n\lambda$ C1
for $\theta = 90°$
$1 / (650 \times 10^{3}) = n \times 590 \times 10^{-9}$ M1
$n = 2.6$
number of orders is 2 A1 [3]

(d) intensity / brightness decreases (as order increases) B1 [1]

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4 **(a)** when waves overlap / meet, (resultant) displacement is the sum of the individual displacements B1 [1]

(b) **(i)** two (ball-type) dippers (M1)
connected to the same vibrating source /motor (A1)
or
one wave source described (M1)
with two slits (A1) [2]

(ii) lamp with viewing screen on opposite side of tank B1
means of freezing picture e.g. strobe B1 [2]

(c) **(i)** two correct lines labelled X B1 [1]

(ii) correct line labelled N B1 [1]

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5 (a) (i) to produce coherent sources or constant phase difference — B1 [1]

(ii) 1. 360° / 2π rad allow n × 360° or n × 2π (unit missing –1) — B1 [1]

2. 180° / π rad allow (n × 360°) – 180° or (n × 2π) – π — B1 [1]

(iii) 1. waves overlap / meet — B1
(resultant) displacement is sum of displacements of each wave — B1 [2]

2. at P crest on trough (OWTTE) — B1 [1]

(b) $\lambda = ax / D$ — C1
$= 2 \times 2.3 \times 10^{-3} \times 0.25 \times 10^{-3} / 1.8$ — C1
= 639 nm — A1 [3]

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6 (a) waves overlap — B1
(resultant) displacement is the sum of the displacements of each of the waves — B1 [2]

(b) waves travelling in opposite directions overlap / incident and reflected waves overlap
(allow superpose or interfere for overlap here) — B1
waves have the same speed and frequency — B1 [2]

(c) (i) time period = 4 × 0.1 (ms) — C1
$f = 1 / T = 1 / 4 \times 10^{-4} = 2500\,\text{Hz}$ — A1 [2]

(ii) 1. the microphone is at an antinode and goes to a node and then an antinode / maximum amplitude at antinode and minimum amplitude at node — B1 [1]

2. $\lambda / 2 = 6.7\,(\text{cm})$ — C1
$v = f\lambda$ — C1
$v = 2500 \times 13.4 \times 10^{-2} = 335\,\text{m s}^{-1}$ — A1 [3]
incorrect λ then can only score second mark

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7 (a) (i) coherence: constant phase difference — M1
between (two) waves — A1 [2]

(ii) path difference is *either* λ *or* $n\lambda$
or phase difference is 360° *or* $n \times 360°$ *or* $n2\pi$ rad — B1 [1]

(iii) path difference is *either* $\lambda/2$ *or* $(n + ½)\,\lambda$
or phase difference is odd multiple of *either* 180° *or* π rad — B1 [1]

(iv) $w = \lambda D / a$ — C1
$= [630 \times 10^{-9} \times 1.5] / 0.45 \times 10^{-3}$ — C1
$= 2.1 \times 10^{-3}\,\text{m}$ — A1 [3]

(b) no change to <u>dark</u> fringes — B1
no change to separation/fringe width — B1
<u>bright</u> fringes are brighter/lighter/more intense — B1 [3]

8	**(a) (i)**	diffraction bending/spreading of light at edge/slit	B1	
		this occurs at each slit	B1	[2]
	(ii)	constant phase difference between each of the waves	B1	[1]
	(iii)	(when the waves meet) the resultant displacement is the sum of the displacements of each wave	B1	[1]
	(b)	$d\sin\theta = n\lambda$		
		$n = d / \lambda = 1 / 450 \times 103 \times 630 \times 10^{-9}$	C1	
		$n = 3.52$	M1	
		hence number of orders = 3	A1	[3]
	(c)	λ blue is less than λ red	M1	
		more orders seen	A1	
		each order is at a smaller angle than for the equivalent red	A1	[3]

9	**(a)**	waves pass through the elements / gaps / slits in the grating	M1	
		spread into geometric shadow	A1	[2]
	(b) (i) 1.	displacements add to give resultant displacement	B1	
		each wavelength travels the same path difference or are in phase	B1	
		hence produce a maximum	A0	[2]
	2.	to obtain a maximum the path difference must be λ or phase difference 360° / 2π rad	B1	
		λ of red and blue are different	B1	
		hence maxima at different angles / positions	A0	[2]
	(ii)	$n\lambda = d \sin\theta$	C1	
		$N = \sin 61° / (2 \times 625 \times 10^{-9}) = 7.0 \times 10^{5}$	A1	[2]
	(iii)	$n\lambda = 2 \times 625$ is a constant (1250)	C1	
		$n = 1 \rightarrow \lambda = 1250$ outside visible		
		$n = 3 \rightarrow \lambda = 417$ in visible		
		$n = 4 \rightarrow \lambda = 312.5$ outside visible		
		$\lambda = 420$ nm	A1	[2]

10	**(a)**	waves overlap / meet / superpose	(B1)	
		coherence / constant phase difference (*not constant λ or frequency)*	(B1)	
		path difference = 0, λ, 2λ or phase difference = 0, 2π, 4π	(B1)	
		same direction of polarisation/unpolarised	max. 3 (B1)	[3]
	(b)	$\lambda = v / f$	C1	
		$f = 12 \times 10^{9}$ Hz	C1	
		$\lambda = 3 \times 10^{8} / 12 \times 10^{9}$ (*any subject)*	M1	
		$= 0.025$ m	A0	[3]
	(c)	maximum at P	B1	
		<u>several</u> minima or maxima between O and P	B1	
		5 maxima / 6 minima between O and P or 7 maxima / 6 minima including O and P	B1	[3]
	(d)	slits made narrower	B1	
		slits put closer together	B1	[2]
		(*not just 'make slits smaller')*		
		Allow tilting the slits M1 and explanation of axes of rotation A1		

11	**(a)**	when waves overlap / meet	B1	
		the resultant displacement is the sum of the individual displacements of the waves	B1	[2]
	(b) (i)	**1.** phase difference = 180° / (n + ½) 360° (allow in rad)	B1	[1]
		2. phase difference = 0 / 360° / (n360°) (allow in rad)	B1	[1]
	(ii)	$v = f\lambda$	C1	
		λ = 320 / 400 = 0.80 m	A1	[2]
	(iii)	path difference = 7 – 5 = 2 (m)		
		= 2.5λ	M1	
		hence minimum		
		or maximum if phase change at P is suggested	A1	[2]
12	**(a)**	<u>waves</u> from the double slit are coherent / constant phase difference	B1	
		<u>waves</u> (from each slit) overlap / superpose / meet (**not** interfere)	B1	
		maximum / bright fringe where path difference is $n\lambda$		
		or phase difference is n360° / 2πn rad		
		or minimum / dark fringe where path difference is $(n + \frac{1}{2})\lambda$		
		or phase difference is (2n + 1) 180° / (2n + 1)π rad	B1	[3]
	(b)	$v = f\lambda$	C1	
		$\lambda = (3 \times 10^8) / 670 \times 10^{12}$ = 448 (or 450) (nm)	M1	[2]
	(c)	w = 12 / 9	C1	
		a (= $D\lambda / w$) = $(2.8 \times 450 \times 10^{-9}) / (12 / 9 \times 10^{-3})$ [allow nm, mm]	C1	
		= 9.5×10^{-4} m [9.4×10^{-4} m using λ = 448 nm]	A1	[3]
	(d)	(red light has) larger / higher / longer wavelength (must be comparison)	M1	
		fringes further apart / larger separation	A1	[2]
13	**(a)**	difference: vibration / oscillation (of particles) / displacement of particles is parallel to energy transfer / wavefronts in longitudinal and perpendicular for transverse	B1	
	or	transverse can be polarised, longitudinal cannot be polarised		
		similarity: both transfer / propagate energy	B1	[2]
	(b) (i)	waves from <u>slits</u> are coherent / constant phase relationship	(B1)	
		waves overlap (at screen) with a phase difference or have a path difference	(B1)	
		maxima where phase difference is integer ×360° (or ×2π rad)		
		or path difference is integer ×λ		
		or equivalent explanation of minima e.g. (n+½)×360°	(B1)	
		max. 2		[2]
	(ii)	maxima spacing = λD / a	C1	
		= $(6.3 \times 10^{-7} \times 2.5) / 0.35 \times 10^{-3}$		
		= 4.5×10^{-3} m	A1	[2]
	(c)	(ultra-violet has) short<u>er</u> wavelength, hence small<u>er</u> separation / distance	A1	[1]

14	**(a)**		diffraction is the spreading of a wave as it passes through a slit or past an edge	B1	
			when two (or more) waves superpose/meet/overlap	M1	
			resultant displacement is the sum of the displacement of each wave	A1	[3]
	(b)		$n\lambda = d \sin \theta$ and $v = f\lambda$	C1	
			max order number for $\theta = 90°$		
			hence n (= f/vN) = $7.06 \times 10^{14} / (3 \times 10^8 \times 650 \times 10^3)$	M1	
			$n = 3.6$		
			hence number of orders = 3	A1	[3]
	(c)		greater wavelength so fewer orders seen	A1	[1]

15	**(a)**	**(i)**	coherent: constant phase difference	B1	
			interference is the (overlapping of waves and the) sum of/addition of displacement of two waves	B1	[2]
		(ii)	wavelength = 3.2 m (allow ± 0.05 m)	M1	
			f (= v/λ = 240 / 3.2) = 75 Hz	A1	[2]
		(iii)	90° (allow ± 2°) or $\pi/2$ rad	A1	[1]
		(iv)	sketch has amplitude 3.0 ± 0.1 cm	M1	
			correct displacement values at previous peaks to produce correct shape	A1	[2]
	(b)	**(i)**	$\lambda = ax/D$	C1	
			$x = (546 \times 10^{-9} \times 0.85) / 0.13 \times 10^{-3}$ (= 3.57×10^{-3} m)	C1	
			AB = 8.9 (8.93) $\times 10^{-3}$ m	A1	[3]
		(ii)	shorter wavelength for blue light so separation is less	B1	[1]

16	**(a)**	**(i)**	Displacement of particles perpendicular to direction of energy propagation	B1	
		(ii)	wave<u>s</u> meet / overlap (at a point)	B1	
			(resultant) displacement is sum of the individual displacements	B1	
	(b)	**(i)**	$\lambda = vT$ **or** $\lambda = v/f$ and $f = 1/T$	C1	
			$\lambda = 4.0 \times 1.5$		
			$\lambda = 6.0$ (cm)	A1	

17		**(i)**	gradient = $(4.5 - 2.4) \times 10^{-3} / (3.25 - 1.75)$ [= 1.4×10^{-3}		
			wavelength = $0.45 \times 10^{-3} \times 1.4 \times 10^{-3}$	C1	
			= 6.30×10^{-7} (m)	C1	
			= 630 nm	A1	[4]
		(ii)	(gradient is equal to λ/a therefore) gradient of line is reduced	B1	
			value of x will be reduced for all values of D *or* new line is completely below old line *or* intercept is less	B1	[2]

18 (a) diffraction: spreading/diverging of waves/light (takes place) at (each) slit/ element/gap/aperture	B1	
interference: overlapping of waves (from coherent sources at each element)	B1	
path difference λ/phase difference of 360(°)/2π (produces the first order)	B1	[3]
(b) $d \sin\theta = n\lambda$ or $\sin\theta = Nn\lambda$	C1	
$d = (2 \times 486 \times 10^{-9}) / \sin 29.7°$ (= 1.962×10^{-6})	C1	
number of lines = 510 (509.7) mm^{-1}	A1	[3]
19 (a) wave incident on/passes by or through an aperture/edge	B1	
wave spreads (into geometrical shadow)	B1	[2]
(b) $n\lambda = d \sin\theta$	C1	
substitution of $\theta = 90°$ *or* $\sin\theta = 1$	C1	
$4 \times 500 \times 10^{-9} = d \times \sin 90°$		
line spacing = 2.0×10^{-6} m	A1	[3]
(c) wavelength of red light is long<u>er</u> (than 500 nm)	M1	
(each order/fourth order is now at a greater angle so) the fifth-order maximum cannot be formed/not formed	A1	[2]
20 (a) wave incident on/passes by or through an aperture/edge	B1	
wave spreads (into geometrical shadow)	B1	[2]
(b) (i) waves (from slits) overlap (at point X)	B1	
path difference (from slits to X) is zero/ phase difference (between the two waves) is zero (so constructive interference gives bright fringe)	B1	[2]
(ii) difference in distances = $\lambda/2 = 580/2$ = 290 nm	A1	[1]
(iii) $\lambda = ax/D$	C1	
$D = [0.41 \times 10^{-3} \times (2 \times 2.0 \times 10^{-3})] / 580 \times 10^{-9}$	C1	
= 2.8 m	A1	[3]
(iv) same separation/fringe width/number of fringes bright fringe(s)/central bright fringe/(fringe at) X less bright dark fringe(s)/(fringe at) Y/(fringe at) Z brighter contrast between fringes decreases *Any two of the above four points, 1 mark each*	B2	[2]

21 (a) (i)	<u>waves</u> at (each) slit/aperture spread	**B1**
	(into the geometric shadow) <u>wave(s)</u> overlap/superpose/sum/meet/intersect	**B1**
(ii)	there is not a constant phase difference/coherence (for two separate light source(s))	**B1**
or	waves/light from the double slit are coherent/have a constant phase difference	
(b)	$x = \lambda D / a$	**C1**
	$\lambda = (36 \times 10^{-3} \times 0.48 \times 10^{-3}) / (16 \times 2.4)$	**C1**
	$= 4.5 \times 10^{-7}$ m	**A1**
(c) (i)	<u>no</u> movement of the water/water is flat/no ripples/disturbance	**B1**
	the path difference is 2.5λ **or** the phase difference is 900° **or** 5π rad	**B1**
(ii) 1.	surface/water/P vibrates/ripples **and** as (waves from the two dippers) arrive in phase	**B1**
2.	surface/water/P vibrates/ripples **and** as amplitudes/displacements are no longer equal/do not cancel	**B1**

22 (a) (i)	waves spread at (each) slit/gap	**B1**
(ii)	constant phase difference (between (each of) the waves)	**B1**
(b) (i)	$n\lambda = d\sin\theta$	**B1**
	$d\sin\theta$ is the same **and** $3\lambda_1 = 4\lambda_2$ so $\lambda_2 / \lambda_1 = 0.75$	**A1**
(ii)	$\lambda_2 / \lambda_1 = 0.75$ and $\lambda_1 - \lambda_2 = 170$	**A1**
	$\lambda_1 = 680$ nm	

23 (a)	intensity $\propto$ (amplitude)^2	**B1**
(b) (i)	$v = f\lambda$ **or** $c = f\lambda$	**C1**
	$f = 3.00 \times 10^8 / 0.060$	**A1**
	$= 5.0 \times 10^9$ Hz	
(ii)	(at X path) difference $= 3\lambda$	**M1**
	(at X phase) difference = 0 or 1080°	**M1**
	so intensity is at a maximum/it is an intensity maximum	**A1**
(iii) 1.	decrease in the distance between (adjacent intensity) maxima/minima	**B1**
2.	(intensity) maxima and minima exchange places	**B1**

24 (a) when (two or more) waves meet (at a point) **B1**

(resultant) displacement is the sum of the individual displacements **B1**

(b)(i) constant phase difference (between the waves) **B1**

(ii) **1.** phase difference = 360° **or** 0 **B1**

2. path difference $= 1.5\lambda$ **A1**

$= 1.5 \times 610$

$= 920\,\text{nm}$

(iii) $\lambda = ax/D$ **C1**

$x = 22/4$ (= 5.5 mm) **or** $22 \times 10^{-3}/4$ ($= 5.5 \times 10^{-3}$ m) **C1**

$a = (610 \times 10^{-9} \times 2.7)/(5.5 \times 10^{-3})$ **A1**

$= 3.0 \times 10^{-4}\,\text{m}$

(iv) shorter wavelength **and** (so) separation decreases **B1**

(v) **B2**

- no change to fringe separation/fringe width/number of fringes
- bright fringes are brighter
- dark fringes are unchanged

Any two of the above three points, 1 mark each.

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25 (a) $n\lambda = d\sin\theta$ **C1**

$\lambda = 640 \times 10^{-9}$ (m) **C1**

$2 \times 640 \times 10^{-9} = 1.7 \times 10^{-6} \times \sin\theta$ so $\theta = 49(°)$ **A1**

(b) $2 \times 640 \times 10^{-9} = 3 \times \lambda$ **C1**
or
$1.7 \times 10^{-6} \times \sin 49° = 3 \times \lambda$

$\lambda = 4.3 \times 10^{-7}\,\text{m}$ **A1**

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26 (a)	graph with x-axis labelled 'distance' and wavelength/λ correctly shown	**B1**
	graph with x-axis labelled 'time' and period/T correctly shown	**B1**
	graph with y-axis labelled 'displacement' and amplitude/A correctly shown	**B1**
(b)(i)	wave (moves along string and) reflects at fixed point/Y/X/end/wall/boundary	**B1**
	the incident and reflected waves interfere/superpose	**B1**
(ii)	100/40 **or** 2.5 (cycles/periods/T)	**C1**
	1. displacement = 0	**B1**
	2. distance = 130 mm	**A1**
(iii)	**1.** $f = 1/40 \times 10^{-3}$	**A1**
	$= 25\,\text{Hz}$	
	2. $v = f\lambda$ **or** $\lambda = vT$	**C1**
	$\lambda = 30/25$ **or** $30 \times 40 \times 10^{-3}$ (= 1.2 m)	**C1**
	distance $= 1.2 \times 1.5$	**A1**
	$= 1.8\,\text{m}$	

27 (a) (i)	**1.** $N\lambda$	**B1**
	2. N/f	**B1**
(ii)	v (= distance / time) $= N\lambda / (N/f)$ so $v = f\lambda$	**B1**
(b)	$T = 4.0 \times 0.20 = 0.80$ (ms) **or** 8.0×10^{-4} (s)	**C1**
	$f = 1/8.0 \times 10^{-4}$	**A1**
	$= 1300\,\text{Hz}$	
(c) (i)	constant phase difference (between the waves)	**B1**
(ii)	180°	**A1**
(iii)	path difference $= 2\lambda$ **or** $S_1Y - S_2Y = 2\lambda$	**C1**
	distance $= 7.40 + (0.85 \times 2)$	**A1**
	$= 9.1\,\text{m}$	

28 (a)	(resultant) force proportional/equal to/is rate of change of momentum	**B1**
(b)(i)	distance $= $ area under graph **or** $s = ½ (u + v) t$	**C1**
	$= ½ \times (9 + 13) \times 10$	
	or	
	$s = ut + ½at^2$	
	$= (9 \times 10) + (½ \times 0.40 \times 10^2)$	
	or	
	$s = vt – ½at^2$	
	$= (13 \times 10) – (½ \times 0.40 \times 10^2)$	
	or	
	$v^2 = u^2 + 2as$	
	$13^2 = 9^2 + (2 \times 0.40 \times s)$	
	distance $= 110$ m	**A1**
(ii)	**1.** $a =$ gradient **or** $a = (v – u) / t$ **or** $a = \Delta v / (\Delta)t$	**C1**
	e.g. $a = (14 – 9) / 12.5$ **or** $(13 – 9) / 10$	
	$a = 0.40$ m s$^{–2}$	**A1**
	2. resultant force $= 850 \times 0.40$	**A1**
	$= 340$ N	
	3. ($F =$) $510 + 440 + 340 = 1300$ (N)	**A1**
	4. $P = Fv$	**C1**
	$= 1300 \times 13$	**A1**
	$= 1.7 \times 10^4$ W	
(c)	$E = \sigma / \varepsilon$	**C1**
	$E = (F / A) / (\Delta L / L)$ **or** $E = FL / A\Delta L$	**C1**
	$\Delta L = (480 \times 0.48) / (3.0 \times 10^{–4} \times 2.2 \times 10^{11})$	**A1**
	$= 3.5 \times 10^{–6}$ m	
(d)	$f_o = f_s v / (v – v_s)$	**C1**
	$480 = f_s \times 340 / (340 – 14)$	
	$f_s = 460$ Hz	**A1**

29.

4(a)(i)	distance (in a specified direction of particle/point on wave) from the equilibrium position	**B1**
4(a)(ii)	the maximum distance (of particle/point on wave) from the equilibrium position **or** the maximum displacement (of particle/point on wave)	**B1**
4(b)	$I \propto A^2$	**C1**
	$I_R / I = (3.6 - 1.2)^2 / (1.2)^2$ resultant intensity = $4.0I$	**A1**
4(c)(i)	as wave(s) pass through the slit(s)	**B1**
	wave(s) spread (into geometric shadow)	**B1**
4(c)(ii)	$n\lambda = d \sin \theta$	**C1**
	$3\lambda = d \sin 90°$ **or** $3\lambda = d$	**C1**
	$d = 3 \times 630 \times 10^{-9}$ $= 1.9 \times 10^{-6}$ m	**A1**
4(c)(iii)	wavelength of blue light is shorter (than 540 nm/630 nm/wavelengths of original light)	**M1**
	(so) third order diffraction maximum is produced	**A1**

30.

5(a)	(incident) wave reflects at end/top of tube	**B1**
	(incident) wave and reflected wave interfere/superpose	**B1**
5(b)	line has maximum value of amplitude at $h = 0$ and $h = 0.60$ m only	**B1**
	line has minimum/zero value of amplitude at $h = 0.30$ m only	**B1**
5(c)(i)	vertical/along length of tube/along axis of tube	**B1**
5(c)(ii)	phase difference = 0	**A1**
5(d)	$v = f\lambda$	**C1**
	$v = 340 / (2 \times 0.60)$ $= 280$ Hz	**A1**
5(e)	$f = 340 / 0.60$ $= 570$ Hz	**A1**

31.

5(a)(i)	the dippers are connected to the same vibrator/motor	**B1**
5(a)(ii)	(the overlapping waves have) similar/same amplitude	**B1**
5(b)	any means of ‘freezing’ the pattern e.g. use a stroboscope/strobe	**B1**
5(c)	$vT = \lambda$ **or** $v = f\lambda$ **and** $f = 1 / T$	**C1**
	$T = 0.060 / 0.40$ $= 0.15$ s	**A1**
5(d)(i)	path difference = 3.0 cm	**A1**
5(d)(ii)	phase difference = 180°	**A1**
5(e)	line drawn joining points where only maxima are observed (i.e. through points where wavefronts intersect) of length at least 4 cm	**B1**

32.

5(a)(i)	(coherence means) constant phase difference (between waves)	**B1**
5(a)(ii)	(interference is) the sum/addition/combination of the displacements of overlapping/meeting waves	**B1**
5(b)(i)	$n\lambda = d \sin\theta$	**C1**
	$\lambda = \sin 51° / (2 \times 6.7 \times 10^{5})$ $= 5.8 \times 10^{-7}$ m	**A1**
5(b)(ii)	smaller angle (corresponding to second order maxima and so) shorter distance (between second order maxima spots)	**B1**

33.

4(a)(i)	vibrations (of particles) are parallel to direction of energy propagation	**B1**
4(a)(ii)	waves meet/overlap (at a point)	**B1**
	(resultant) displacement is sum of individual displacements	**B1**
4(b)(i)	$\lambda = ax / D$	**C1**
	$= (3.7 \times 10^{-4} \times 4.3 \times 10^{-3}) / 2.3$	**C1**
	$= 6.9 \times 10^{-7}$ (m) = 690 nm	**A1**
4(b)(ii)	• no change to fringe separation/fringe width/number of fringes • bright fringes are darker • dark fringes are brighter *Any two marking points, 1 mark each*	**B2**

34.

4(a)	progressive waves transfer energy **or** stationary waves do not transfer energy	**B1**
4(b)(i)	0.32 m	**A1**
4(b)(ii)	$v = \lambda / T$ **or** $v = f\lambda$ **and** $f = 1 / T$	**C1**
	$v = 0.32 / 0.020$ **or** 50×0.32 $= 16$ m s^{-1}	**A1**
4(b)(iii)	450° **or** 90°	**A1**
4(b)(iv)	(P has) maximum downward displacement at 0.005 s	**B1**
	returns to original position/point (at 0.010 s)	**B1**
4(c)(i)	(position where) zero amplitude	**B1**
4(c)(ii)	2	**A1**
4(c)(iii)	180°	**A1**
4(c)(iv)	string drawn between X and Y with one antinode midway along the string	**B1**

35.

6(a)	the waves (of the same type) move in opposite directions and overlap	**B1**
	the waves have the same (speed and) frequency/wavelength	**B1**
6(b)(i)	zero amplitude	**B1**
6(b)(ii)	distance $= 6.0 \times 4$ $= 24$ cm	**A1**
6(b)(iii)	180°	**A1**

36.

5(a)	$v = f\lambda$ **or** $c = f\lambda$	**C1**
	$f = 3.0 \times 10^8 / 0.040$	**C1**
	$= 7.5 \times 10^9$ (Hz) $= 7.5$ GHz	**A1**
5(b)(i)	path difference = 0.020 m	**A1**
5(b)(ii)	phase difference = 180°	**A1**
5(c)	(intensity) increases	**C1**
	(intensity) increases by a factor of 4	**A1**
5(d)(i)	minimum moves closer to the maximum **or** decrease in separation of maximum and minimum	**B1**
5(d)(ii)	the maximum and minimum exchange places **or** the maximum becomes a minimum **and** the minimum becomes a maximum	**B1**

37.

5(a)(i)	$T = 2.0 \times 10^{-5} \times 6.0$ $(= 1.2 \times 10^{-4}$ s)	**C1**
	$f = 1 / (2.0 \times 10^{-5} \times 6.0)$ $= 8300$ Hz	**A1**
5(a)(ii)	new trace shows the same period	**B1**
	new trace shows amplitude of 10 small squares	**B1**
5(a)(iii)	(trace is a) vertical line	**B1**
5(b)(i)	$n\lambda = d \sin\theta$	**C1**
	$\lambda = (3.4 \times 10^{-6} \times \sin 16°) / 2$ $= 4.7 \times 10^{-7}$ m	**A1**
5(b)(ii)	$n = 3.4 \times 10^{-6}$ $(\times \sin 90°) / 4.7 \times 10^{-7}$ or 2 $(\times \sin 90°) / \sin 16°$ (= 7.2 or 7.3)	**C1**
	highest order = 7	**A1**

38.

4(a)	(two or more) waves meet (at a point)	**B1**
	(resultant) displacement is the sum of the individual displacements	**B1**
4(b)(i)	it is a (wave) reflector / it reflects (the wave)	**B1**
4(b)(ii)	$v = f\lambda$ or $c = f\lambda$	**C1**
	$f = 3.0 \times 10^8 / 0.040$ $= 7.5 \times 10^9$ (Hz) $= 7.5 \times 10^9 / 10^9$ (GHz)	**C1**
	= 7.5 GHz	**A1**
4(b)(iii)	1 distance = 0.020 m	**A1**
	2 number = 5	**A1**

39.

4(a)	distance moved by wavefront/energy during one cycle/oscillation/period (of source) **or** minimum distance between two wavefronts **or** distance between two adjacent wavefronts	**B1**
4(b)	$v = \lambda / T$ **or** $v = f\lambda$ **and** $f = 1 / T$	**C1**
	$T = 460 \times 10^{-9} / 3.00 \times 10^8$	**C1**
	$= 1.5 \times 10^{-15}$ s	**A1**
4(c)	waves pass through/enter the slit(s)	**B1**
	waves spread (into geometric shadow)	**B1**
4(d)(i)	$n\lambda = d \sin\theta$	**C1**
	$G = \sin\theta / \lambda$ $d = 4 / G$	**A1**
4(d)(ii)	straight line from 400 nm to 700 nm that is always below printed line	**M1**
	straight line has smaller gradient than printed line and is 5 small squares high at wavelength of 700 nm	**A1**

40.

4(a)	time for one oscillation/vibration/cycle **or** time between adjacent wavefronts (passing the same point) **or** shortest time between two wavefronts (passing the same point)	**B1**
4(b)	(when two or more) waves meet/overlap (at a point)	**B1**
	(resultant) displacement is sum of the individual displacements	**B1**
4(c)(i)	microwave(s)	**B1**
4(c)(ii)	$v = \lambda / T$ **or** $v = f\lambda$ **and** $f = 1/T$	**C1**
	$T = 0.040 / 3.00 \times 10^8$	**C1**
	$= 1.33 \times 10^{-10}$ (s) $= 1.33 \times 10^{-10} / 10^{-12}$ (ps) = 130 ps	**A1**

4(c)(iii)	(1.380 – 1.240) / 0.040 = 3.5 **or** 1.380 / 0.040 – 1.240 / 0.040 = 3.5	**A1**
4(c)(iv)	phase difference = 1260° **or** 180°	**A1**
4(c)(v)	(always) zero	**A1**
4(c)(vi)	increase in distance between (adjacent intensity) maxima/minima	**A1**

41.

4(a)	(when two or more) waves meet/overlap (at a point)	**B1**
	(resultant) displacement is sum of the individual displacements	**B1**
4(b)	intensity $\propto$ amplitude^2	**C1**
	maximum intensity = $9I$	**A1**
4(c)(i)	$x = \lambda D / a$	**C1**
	$= (550 \times 10^{-9} \times 1.2) / (0.35 \times 10^{-3})$	**C1**
	$= 1.9 \times 10^{-3}$ m	**A1**
4(c)(ii)	red light has longer wavelength (than 550 nm) so distance (between fringes) increases	**B1**

42.

5(a)	a (much) louder sound can be heard	**B1**
5(b)	$v = f\lambda$	**C1**
	$\lambda = 340 / 530$ (= 0.64 m)	**C1**
	height = 0.64 / 4 = 0.16 m	**A1**
5(c)	distance = 0.64 / 2 **or** 0.16 × 2 **or** (¾ × 0.64 – ¼ × 0.64) = 0.32 m	**A1**

43.

5(a)	maximum displacement (of a point/particle on string/wave)	**B1**
5(b)	$v = \lambda / T$ **or** $v = f\lambda$ **and** $f = 1 / T$	**C1**
	$T = 690 \times 10^{-9} / 3.00 \times 10^{8}$	**C1**
	$= 2.3 \times 10^{-15}$ s	**A1**
5(c)(i)	$\lambda = ax / D$	**C1**
	$G = x / D$ (so) $a = \lambda / G$	**A1**
5(c)(ii)	straight line from origin always below printed line	**M1**
	line is half the height of printed line at maximum D	**A1**

44.

5(a)	wave(s) (travel along string and) reflect at fixed point / A / B / end	**B1**
	incident and reflected waves superpose / interfere **or** two waves travelling / with speed in opposite directions superpose / interfere	**B1**
5(b)	line has an approximate sinusoidal shape with maximum downward displacement at P and zero displacement at each node	**B1**
5(c)	$v = \lambda / T$ **or** $v = f\lambda$ **and** $f = 1 / T$	**C1**
	$\lambda = 35 \times 0.040$ **or** $35 / 25$ $(= 1.4\ \text{m})$	**C1**
	distance = 1.4×2.5 = 3.5 m	**A1**
5(d)	(number of periods / cycles) (= t / T) = 0.060 / 0.040 (= 1.5)	**C1**
	amplitude = 72 / 6 = 12 mm	**A1**

45.

5(a)	constant phase difference (between the waves)	**B1**
5(b)(i)	path difference = 1.5×660 = 990 nm	**A1**
5(b)(ii)	phase difference = $360° \times 1.5$ = 540°	**A1**
5(c)	$\lambda = ax / D$	**C1**
	$x = (660 \times 10^{-9} \times 1.8) / 0.44 \times 10^{-3}$	**C1**
	$= 2.7 \times 10^{-3}$ m	**A1**
5(d)	bright fringes are brighter	**B1**
	no change to dark fringes	**B1**
	no change to (fringe) separation / (fringe) spacing	**B1**
5(e)	(blue light has) shorter wavelength	**M1**
	(so) decrease (slit) separation	**A1**

46.

5(a)(i)	power = intensity × area	**C1**
	$= 1.3 \times 10^{3} \times (\pi \times 0.055^{2})$ = 12 W	**A1**
5(a)(ii)	intensity = power / area $= 12 / (\pi \times 0.0015^{2})$ $= 1.7 \times 10^{6}\ \text{W m}^{-2}$	**A1**
5(b)(i)	(λ =) v / f **or** c / f	**C1**
	(λ =) $3.0 \times 10^{8} / 3.7 \times 10^{15} = 8.1 \times 10^{-8}$ (m)	**A1**
5(b)(ii)	ultraviolet	**A1**

5(b)(iii)	$d \sin \theta = n\lambda$ **or** $(1 / N) \times \sin \theta = n\lambda$	**C1**
	$d = 1 / 2400 \times 10^3$ (m) $= 4.2 \times 10^{-7}$ (m) **or** $N = 2400 \times 10^3$ (m^{-1})	**C1**
	$n = 4.2 \times 10^{-7} \times \sin 90° / 8.1 \times 10^{-8}$ **or** $\sin 90° / 2400 \times 10^3 \times 8.1 \times 10^{-8}$ $n = 5.2$ or 5.1 **or** when $n = 5$, $\theta = 76.4°$ and when $n = 6$, $\sin\theta > 1$ (so) $n = 5$	**B1**
	number of maxima = $(2 \times 5) + 1$ $= 11$	**A1**
5(b)(iv)	the wavelength has increased	**M1**
	(so) number of maxima decreases	**A1**

47.

5(a)(i)	period or $T = 1 / 5000$ ($= 2 \times 10^{-4}$ s)	**C1**
	time-base setting $= 1.5 \times 2 \times 10^{-4} / 6.0$ **or** $2 \times 10^{-4} / 4.0$ $= 5 \times 10^{-5}$ s cm^{-1}	**A1**
5(a)(ii)	new trace drawn with same period as original trace	**B1**
	new trace drawn with amplitude greater than 1.0 cm	**M1**
	new trace drawn with amplitude of 1.7 cm	**A1**
5(b)(i)	path difference (from slits to P) is zero **or** phase difference (between waves at P) is zero (so constructive interference)	**B1**
5(b)(ii)	$\lambda = ax / D$	**C1**
	$D = (3.6 \times 10^{-4} \times 4.0 \times 10^{-3}) / 630 \times 10^{-9}$	**C1**
	$= 2.3$ m	**A1**
5(c)	upward sloping straight line starting from a non-zero value of x at $\lambda = 400$ nm	**B1**

48.

5(a)(i)	infrared	**B1**
5(a)(ii)	$v = f\lambda$ **or** $c = f\lambda$	**C1**
	$f = 3.0 \times 10^8 / 8.4 \times 10^{-6}$ $= 3.6 \times 10^{13}$ (Hz) $= 36$ THz	**A1**
5(b)(i)	constant phase difference (between the waves) (with time)	**B1**
5(b)(ii)	$\lambda = ax / D$	**C1**
	$x = 22 / 8$ **or** 2.75 (mm) **or** $22 \times 10^{-3} / 8$ **or** 2.75×10^{-3} (m)	**C1**
	$a = (6.2 \times 10^{-7} \times 2.8) / (22 \times 10^{-3} / 8)$ $= 6.3 \times 10^{-4}$ m	**A1**

5(c)(i)	difference in distances $= 6.2 \times 10^{-7} / 2$ $= 3.1 \times 10^{-7}$ m	A1
5(c)(ii)	phase difference = 360°	A1

49.

4(a)	the number of wavefronts/crests/troughs passing a fixed point per unit time **or** the number of oscillations per unit time (of source / point on wave / particle of medium)	B1
4(b)(i)	$T = 4 \times 0.50 \times 10^{-3}$ $(= 2.0 \times 10^{-3}$ s)	C1
	$f = 1 / 2.0 \times 10^{-3}$ = 500 Hz	A1
4(b)(ii)	amplitude $= 2.8 \times 0.20$ = 0.56 V	A1
4(c)	period same as original trace	B1
	sinusoidal wave of constant amplitude less than 2.8 cm throughout	M1
	amplitude 1.4 cm	A1
4(d)(i)	when (two or more) waves meet (at a point)	B1
	(resultant) displacement is the sum of the individual displacements	B1
4(d)(ii)	node-to-node separation is $\lambda / 2$ **or** microphone moves through 3 node-to-node separations **or** $d = 1.5\lambda$	C1
	$\lambda = 1.05 / 1.5$ = 0.70 m	A1
4(d)(iii)	$v = f \times \lambda$	C1
	$= 500 \times 0.70$ $= 350$ m s^{-1}	A1

50.

4(a)	(when two or more) waves meet / overlap (at a point)	B1
	(resultant) displacement is the sum of the individual displacements	B1
4(b)	$\lambda = ax / D$	C1
	$\lambda = [(0.65 \times 10^{-3}) \times (1.7 \times 10^{-3})] / 1.9$	C1
	$\lambda = 5.8 \times 10^{-7}$ m	A1
4(c)(i)	waves are out of phase by a quarter of a cycle / period **or** when one wave has maximum/minimum displacement, the other wave has zero displacement	B1
4(c)(ii)	(statement 2 is) not correct (because) waves do not have phase difference of 180° / not in antiphase **or** one wave has some displacement when other has no displacement **or** the displacements of the waves are not always equal and opposite	B1

4(d)	more diffraction (of light/waves by slits) **or** light/waves are more spread (by slits) **or** light/waves (from slits) have less intensity	**B1**
	more (bright/dark) fringes **or** bright fringes are less bright / are dimmer / have lower intensity **or** no change to fringe spacing/separation/width	**B1**

51.

5(a)(i)	light intensity has maximum value at 0°, 180°, 360° and zero intensity at 90°, 270°	**M1**
	'sinusoidally-shaped' curve	**A1**
5(a)(ii)	$4.2 = 7.6\cos^2\theta$	**C1**
	$\theta = 42°$	**A1**
5(b)	wave passes (through) an aperture **or** wave passes (by / through / around) an edge	**B1**
	wave spreads (into geometrical shadow)	**B1**
5(c)(i)	$n\lambda = d\sin\theta$	**C1**
	$d = (3 \times 4.3 \times 10^{-7}) / \sin 68°$ $= 1.4 \times 10^{-6}$ m	**A1**
5(c)(ii)	$1.4 \times 10^{-6} \times \sin 68° = 2 \times \lambda$ **or** $3 \times 4.3 \times 10^{-7} = 2 \times \lambda$	**C1**
	$\lambda = 6.5 \times 10^{-7}$ m	**A1**

52.

5(a)(i)	cross labelled Y drawn: at any position where wavefronts cross **or** centrally in a 'diamond' shape formed between any adjacent wavefronts from A and B	**B1**
5(a)(ii)	cross labelled Z drawn on a wavefront from one source at a point midway between adjacent wavefronts from the other source	**B1**
5(b)(i)	$\lambda = ax / D$	**C1**
	$a = (2.9 \times 10^{-5} \times 140) / (1.2 \times 10^{-2})$	**C1**
	$= 0.34$ m	**A1**
5(b)(ii)	infrared	**A1**